

Measured inertia and sustained-contact repair

Rigid-body collision model: verified progress and remaining gaps

7 October 2026

Result. The 3D production adapter now accepts authoritative measured mass, center of mass and inertia. A separate supported sphere/disk primitive includes distinct static/dynamic friction and independent rolling/spin angular impulses. It slows zero-slip rolling, arrests axial spin, and resolves transitions without numerical reversal. No published restitution or friction value was changed or fitted.

What this does not establish. The supported primitive is not integrated into the native general-impact/contact-group solver. The 24 glass comparisons retain exactly their prior errors. No new improvement against experimental measurements has been established, and a general authentic 2D/3D impact model remains unqualified. The original symbolic article is preserved separately.

Diagnosis and implemented changes

Gap	Repair or evidence	Remaining scope
Measured mass/COM/inertia rejected	Explicit SI mass-properties input; full tensor mapped to native principal axes	Source must supply real tensor; extent tests are necessary, not a realizability proof
No sustained rolling/spin resistance	Separate force and angular impulse channels; exact static/dynamic branches and arrest events	Scalar central inertia and one supported motion plane
Source density guard could be bypassed by new input	Profile checks authoritative mass and declared homogeneous-sphere inertia	Generic measured input is separate from source-specific assumptions
General directional static closure	Explicit rejection of partial arrest needing torque outside full $\hat{s}/\hat{n}$ span	Requires a specified physical zero/static-direction convention
Impact deformation/history	Previous local-memory and pressure-patch studies remain isolated	Opening, partial stick/slip, yield and group integration not repaired here

Input semantics. Every measured override supplies mass, local COM, and a symmetric body-axis inertia tensor about that COM. Collision geometry stays geometric. Authored world position retains its COM meaning. Finite positive mass, positive inertia eigenvalues, triangle inequalities and a geometry-extent bound are checked. Three legacy scene preparations are bit-identical to revision 512a1b2. The documented glass profile retains its nominal density and homogeneous-sphere assumption; it rejects contradictory overrides.

Evidence classes. Synthetic controls verify mechanics and numerical implementation. Experimental residuals quantify source reproduction under fixed inputs. Native timings measure a scoped kernel cost. These are distinct claims.

Efficient calculation and the user's impulse notation

With a fixed coordinate length ℓ , combined momentum and velocity use

$$\mathbf{V} = \begin{bmatrix} \mathbf{v} \\ \ell \boldsymbol{\omega} \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} \mathbf{p} \\ \mathbf{L}/\ell \end{bmatrix}, \quad \mathbb{M} = \begin{bmatrix} m \mathbb{1} & \mathbb{0} \\ \mathbb{0} & \mathbb{I}/\ell^2 \end{bmatrix}.$$

Small δ denotes physical contact impulses; Δ denotes body changes. The components $\delta p_n, \delta p_t, \delta L_s, \delta L_n$ are scalars:

$$\delta \mathbf{p} = \hat{\mathbf{n}} \delta p_n + \hat{\mathbf{t}} \delta p_t, \quad \delta \mathbf{L} = \hat{\mathbf{s}} \delta L_s + \hat{\mathbf{n}} \delta L_n, \\ \Delta \mathbf{P} = \begin{bmatrix} \delta \mathbf{p} \\ \mathbf{r} \wedge \delta \mathbf{p}/\ell + \delta \mathbf{L}/\ell \end{bmatrix}.$$

Here $\mathbf{r} \wedge$ maps linear impulse to its lever moment in the appropriate dimension; reversal is expressed through its transpose. In 3D, the angular body change is the lever moment *plus* the independent couple. There is no linear s component. In motion, $\hat{\mathbf{t}}$ is full contact-relative velocity divided by its magnitude, and $\hat{\mathbf{s}}$ is full relative angular velocity divided by its magnitude.

For a supported sphere/disk define scalar translation v , rolling angular speed ω_r , radius R , scalar central inertia I , and slip $u = v - R\omega_r$. Tangential force f and independent rolling moment M_r obey

$$\begin{bmatrix} \dot{u} \\ \dot{\omega}_r \end{bmatrix} = \begin{bmatrix} F_d/m \\ 0 \end{bmatrix} + \begin{bmatrix} 1/m + R^2/I & -R/I \\ -R/I & 1/I \end{bmatrix} \begin{bmatrix} f \\ M_r \end{bmatrix}.$$

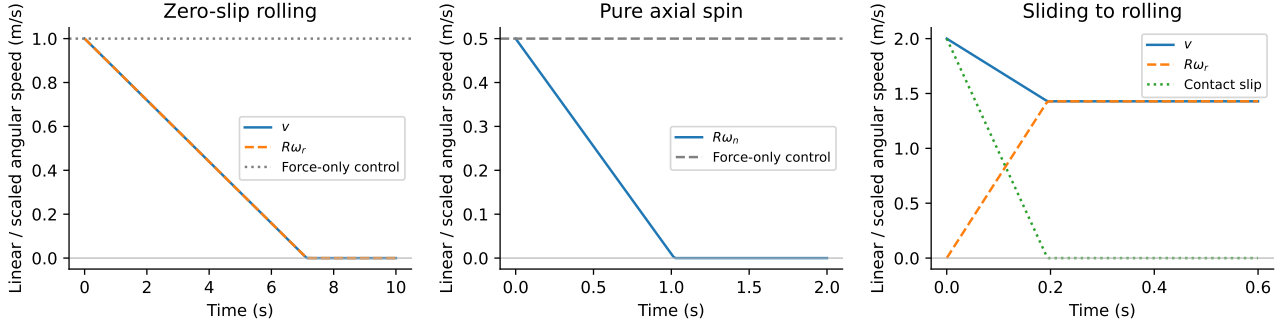
Only two coupled state derivatives are needed. Sliding uses $f = -\mu_d N \text{sign } u$; sticking solves $\dot{u} = 0$ subject to $|f| \leq \mu_s N$. Rolling resistance has capacity $|M_r| \leq \mu_r a_r N$ and opposes nonzero rolling. At zero rolling, the holding reaction is solved subject to this same capacity. Axial spin has declared capacity $|M_n| \leq \mu_n a_n N$ and is integrated only until its arrest.

Constant-branch accelerations are integrated exactly to the earliest arrest, then constraints are re-evaluated. Physical moment lengths a_r, a_n are separate from coordinate length ℓ . Normal/tangential restitution belong to impact branches and are not reapplied in this persistent-load branch.

3D qualification boundary. Rotating this scalar-inertia branch produces $\Delta \mathbf{L} = \mathbf{r} \wedge \delta \mathbf{p} + \delta \mathbf{L}$ with translating-support work included. It does not replace s by a rolling axis. At partial arrest, s can be parallel to n while a transverse static moment is needed: the spatial wrapper explicitly rejects that case. At complete zero motion, static constraint reactions are labeled rather than inventing a velocity unit vector. The independent capacities are a declared phenomenological law, not a measured coupled finite-patch friction budget.

Mechanics controls: motion and spin

Synthetic mechanics controls — no experimental material fit



All plot inputs below are synthetic verification fixtures. Dashed constant-speed controls show why a force-only law at zero slip cannot dissipate rolling/spin. They are not experimental comparisons or material calibration.

Input	Rolling control	Axial-spin control	Slide-to-roll control
m, R, I, N	1 kg, 0.1 m, 0.004 kg m ² , 9.81 N		
v, ω_r, ω_n	1, 10, 0	0, 0, 5	2, 0, 0
μ_s, μ_d	0.5, 0.3	0.5, 0.3	0.8, 0.3
μ_r, a_r	0.02, 0.1 m	0, 0	0, 0
μ_n, a_n	0, 0	0.1, 0.02 m	0, 0

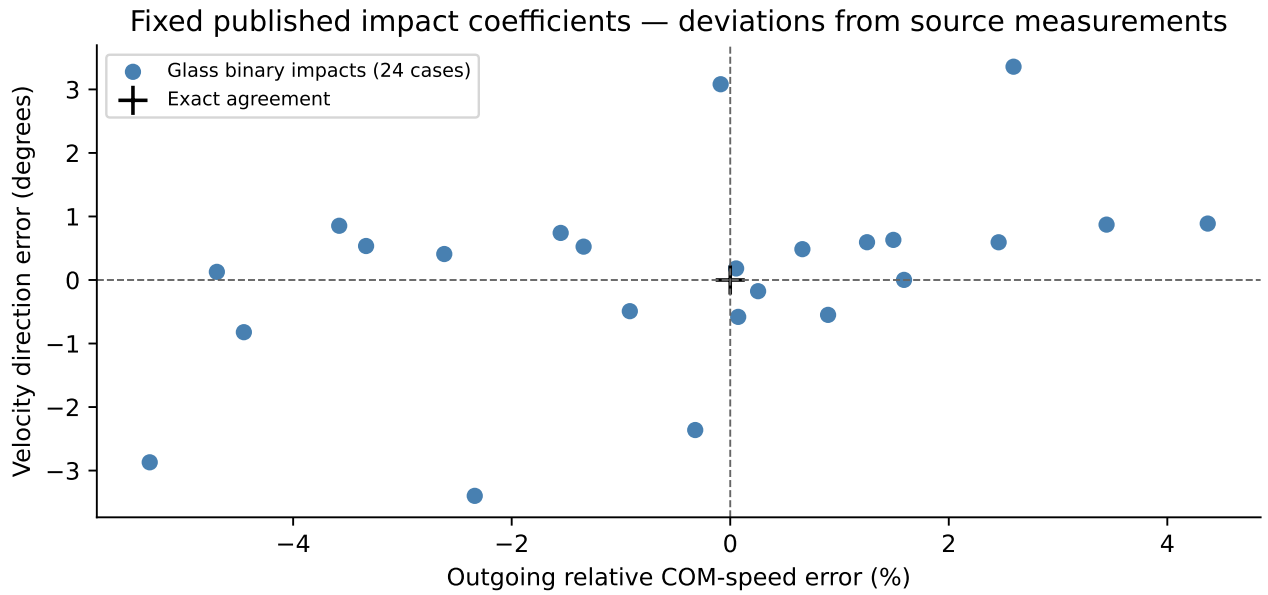
Checks. Fifteen new tests pass, plus 24 existing regression tests. These include static versus dynamic branches, zero-slip rolling, pure axial spin, lever moment plus independent couple, moving-support work, tiny-motion preservation, and explicit rejection of unsupported directional branches.

400 randomized controls pass energy and step-composition checks. Maximum relative energy residual is $1.223\text{e-}15$; maximum scaled composition error is $2.805\text{e-}16$. 100 rotated/moving-support controls have maximum angular-impulse error $2.010\text{e-}15$ N m s. These are finite test samples, not a general error theorem or proof of experimental authenticity.

Native inertia dynamics. A free asymmetric-body rotation test initially missed its 10^{-7} momentum-error tolerance at 64 steps: relative error $3.29 \cdot 10^{-7}$. Refinement to 256 steps gives $8.24 \cdot 10^{-8}$ and passes the unchanged gate. The retained refinement shows first-order gyroscopic integration convergence; no output or material parameter was fitted.

Energy ledger. The supported result records sliding, rolling and axial spin losses separately. For a translating plane with velocity $\mathbf{U}$, support work is $\mathbf{U}^T \delta \mathbf{p}$. Body energy change equals drive work plus support work minus these losses. The new API returns both relative and world energy, preventing a moving support from being mislabeled as contact dissipation.

Comparison to experimental reality



The diagram maps each outgoing *relative COM velocity* to magnitude and direction error for the same incoming worksheet state. The origin is exact agreement. Only one collision family is plotted; no angular-velocity accuracy is implied because the worksheet spin is reconstructed, not independently measured.

24 glass binary impacts	Current result	Comparison status
Normal-speed RMSE	0.036674 m/s	Unchanged from prior comparison
COM-tangent-speed RMSE	0.017805 m/s	Unchanged; source reconstruction retained
Joint input-normalized RMSE	0.024156	Unchanged; dimensionless component RMS
Native analytic/energy checks	24/24	Numerical mechanics checks only

Fixed published impact inputs	Value
Normal restitution e_n	0.97
Tangential restitution e_t	0.44
Pair sliding friction	0.092

The Cornell worksheet and coefficient chart may share characterization trials; this is source reproduction, not an independently calibrated held-out validation. Nominal diameter and homogeneous inertia are source assumptions. Separate static/dynamic/rolling/spin parameters and independent measured spin are absent. Consequently this replay checks compatibility of the native adapter, not the new supported resistance law. All original per-case states and residuals are retained.

Other public-data results remain unchanged: GAUGE sliding forecasts have 6.962/10.824/2.880 mm position RMSE for wood/plastic/metal; its conditional bounce normal-speed RMSE is 0.381 m/s with substantial timing sensitivity. The 134 unique limestone first-rocking ratios have a geometry-only comparator RMSE 0.0390. MIT's 1,718 signed planar states and GAUGE's 160 spatial impact trials are imports/diagnostics, not newly qualified endpoint predictions. These metrics have different units and protocols and must not be pooled into an accuracy score.

Native cost, practical limits and next model work

The warning-free C++17 kernel uses fifteen SoA input arrays and eleven response arrays indexed by body k . It matches all 400 Python fixtures; maximum scaled output difference is zero on this compiler. No fast-math option is used.

Independent updates	ns/update median	Total ms	Array MB
100	48.60	0.005	0.02
10,000	46.94	0.469	2.08
100,000	80.94	8.094	20.80
1,000,000	81.41	81.406	208.00

Five timed repetitions follow one warmup. Input loads, all output stores, branch and energy checks are included. Allocation, collision detection, pose/quaternion evolution, changing normal load, impacts and contact-group solving are excluded. Larger arrays change the memory working set. Repeated independent fixtures are not a million-body interacting simulation. No baseline speedup or $2\times$ gate is claimed. Flat indices and explicit load/store channels are suitable for a later Vektor Flow port; no port or GPU result is claimed here.

What is still missing. General zero/static-direction closure, contact opening and pressure evolution, partial sticking/sliding with shear history, elastic mode energy and appropriate material yield are not captured by this constant-load branch. Arbitrary full-inertia bodies and simultaneous contact groups also remain separate integration tasks. Static/dynamic and angular capacities must eventually be tied to independently characterized contact physics, not just bounded separately because that is easy to implement.

Best next extension. Keep rigid bodies but add a small local contact state for shear displacement and elastic/mode energy, with unilateral normal opening, release of stored energy, static/dynamic transitions, finite-patch force/couple budgets and an auditable work ledger. Start on a single measured oblique impact, then require held-out improvement under fixed published inputs before integrating the branch into general contact groups. The existing patch and memory studies provide components, not a completed constitutive closure.

Maw–Barber–Fawcett’s oblique elastic-sphere analysis supports investigating tangential compliance and partial stick/slip. It does not supply every parameter for rocks or rubber, nor validate this supported-contact primitive. Independently measured mass/inertia, contact timing, endpoint spin, pair-specific friction and contact stiffness/dissipation remain priorities for stronger prediction tests.

Sources and reproduction.

- Cornell original glass [worksheet](#) and [coefficient catalog](#); worksheet SHA-256 in replay summary.
- Maw, Barber, Fawcett (1976), *The oblique impact of elastic spheres*, Wear 38, 101–114. [Author-hosted paper](#), DOI 10.1016/0043-1648(76)90201-5.
- Previous public datasets, protocols, uncertainty and hashes: repository [research/public-validation-data/README.md](#) and its report.
- Current evidence/code: [research/contact-gap-fix/](#); README gives test, audit, replay, benchmark and report commands. Temporary archive relocation is recorded separately; no measurement was deleted.